

Asymptotic and exact sampling distributions of S^2 : Case of Gaussian parent

Statistics 421 - Mathematical Statistics

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Introduction

Understanding the behaviour of the sample variance S^2 is critical for both theoretical studies and practical applications in statistics. This report examines the exact and asymptotic distributions of S^2 from a random sample of Gaussian random variables.

This report utilizes Monte Carlo simulations to verify that the exact distribution of S^2 is $Gamma\left(\frac{n-1}{2}, \frac{2\sigma^2}{n-1}\right)$ (assuming that the sample's parent distribution is $N(\mu, \sigma^2)$). Additionally, to understand how S^2 approximates normality, the report includes a derivation of the asymptotic distribution of S^2 using the Central Limit Theorem (CLT), the Weak Law of Large Numbers (WLLN), and Slutsky's Theorem.

From these findings, this study aims to provide a detailed understanding of the behaviour of S^2 : enhancing the statistical reliability of estimations and hypothesis testing in Gaussian distributions. The findings will bridge theoretical assumptions with practical statistical applications, with a focus on both small and large sample behaviours.

1 Monte Carlo study of S^2

In Statistics, understanding the behaviour of sample variance S^2 under different sampling conditions provides crucial insights for theoretical and applied analytics. This study verifies the exact sampling distribution of S^2 as $Gamma\left(\frac{n-1}{2}, \frac{2\sigma^2}{n-1}\right)$ through Monte Carlo simulations. By generating $R = 10,000$ samples from a normal distribution $N(\mu, \sigma^2)$ with $\mu = 0$ and $\sigma^2 = 1$, and observing sample sizes of $n = 5, 10, 50$, we aim to compare the empirical means and standard deviations (SDs) of S^2 against their theoretical counterparts.

1.a. Analysis Results

The table below summarizes the results from the Monte Carlo simulations:

Table 1: Results from the Monte Carlo Simulations (R=10000)¹

Sample Size (n)	Monte Carlo Mean	Exact Mean	Monte Carlo SD	Exact SD
5	0.9956	1	0.7026	0.7071
10	0.9952	1	0.4730	0.4714
50	0.9972	1	0.2006	0.2020

¹See Appendix A for simulation code

To get a better picture of the closeness, the absolute difference between the Monte Carlo and exact statistics are calculated and summarized in this table:

Sample Size (n)	Difference (Mean)	Difference (SD)
5	0.0044	0.0045
10	0.0048	0.0016
50	0.0028	0.0014

For all sample sizes in the table, the Monte Carlo means and SD's are very close to their exact counterparts; they all differ by less than 0.01. As the sample size increases, the difference between the Monte Carlo mean and the exact mean generally shrinks. Even though there are some exceptions (e.g. $n = 5$ vs $n = 10$ for the mean), the general pattern is still there as the absolute differences get smaller for every other case. These results affirm that as sample size increases, the Monte Carlo simulations provide a more accurate and reliable estimation of both the mean and variability of the sample variance, closely approximating the theoretical Gamma distribution expectations.

Impact of Increasing The Number of Simulations R from 10,000 to 100,000:

The results of changing R from 10,000 to 100,000 in the simulation code and then running it is summarized in this table:

Table 2: Results from the Monte Carlo Simulations ($R=100000$)

Sample Size (n)	Monte Carlo Mean	Exact Mean	Monte Carlo SD	Exact SD
5	1.0004	1	0.7058	0.7071
10	0.9997	1	0.4726	0.4714
50	0.9999	1	0.2017	0.2020

The difference calculations for the simulation is shown in the table below:

Sample Size (n)	Difference (Mean)	Difference (SD)
5	0.0004	0.0013
10	0.0003	0.0012
50	0.0001	0.0003

For all sample sizes, since the differences are smaller, the accuracy of the Monte Carlo means and the Monte Carlo SD's are greater when R is increased to 100,000.

1.b. A Visual Comparison of the Monte Carlo and exact distributions of S^2

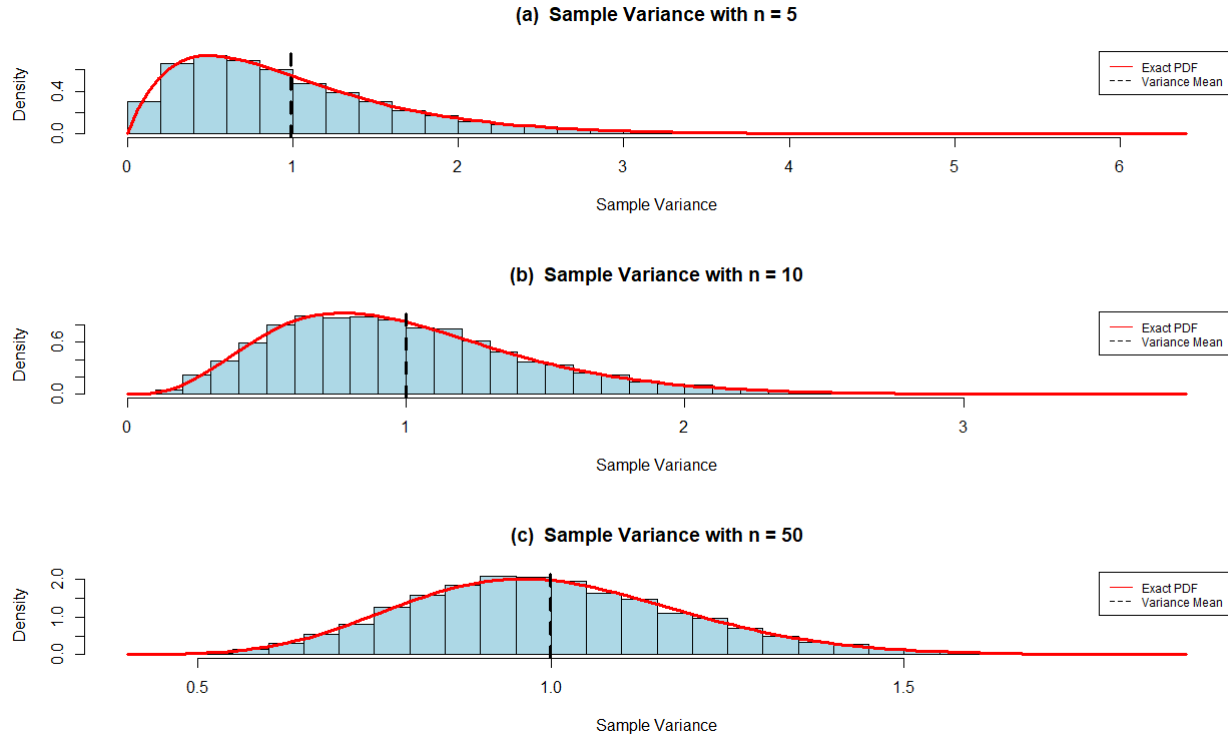


Figure 1: Sample Variance Histograms vs. Exact Gamma PDFs for different sample sizes n , $(R = 10000)^2$

Closeness between Simulated s^2 and Gamma Distribution:

- For $n = 5$, the histogram shows a slight right skew compared to the Gamma PDF, which means the sample variance is greater with smaller sample sizes.
- At $n = 10$, the fit between the histogram and the Gamma PDF improves and the variation is smaller. These effects are even greater when $n = 50$. It is likely a result of the diminished influence of extreme sample values.
- For all sample sizes tested, the shape of the histogram more or less resembles the exact PDF. However, it does seem like a larger sample size reduces their difference.

²See Appendix B for plot code

Impact of Increasing The Number of Simulations R from 10,000 to 100,000:

If `num_simulations` in the code is changed to 100,000, the plots look like this:

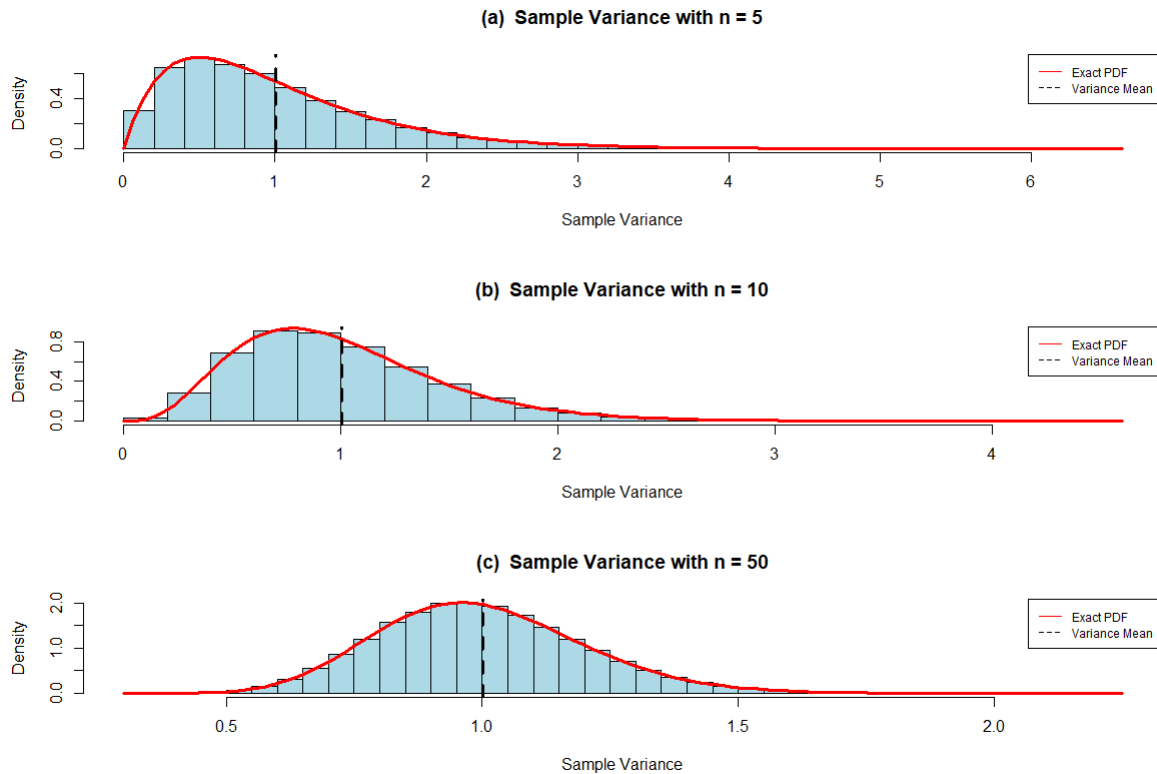


Figure 2: Sample Variance Histograms vs. Exact Gamma PDFs for different sample sizes n , ($R = 100000$)

There is no noticeable change in the closeness when R is increased to 100,000 (Except the time it took for the code to run), which is probably caused by the limitations of the plotting software and the fact that 10,000 simulations is quite a lot. However, the variability of all the simulations should be smaller—as outliers would have less influence on the results. Thus, the more simulations we run, the more the simulated distribution's shape and spread will stabilize, reducing random fluctuations and providing a more accurate comparison to the theoretical distribution.

Conclusions

- Increasing the sample size n leads to a better approximation of the theoretical gamma distribution by the empirical distributions of the sample variance. This is because larger sample sizes help mitigate the effects of random sampling variability, enhancing the robustness and accuracy of statistical estimations.
- Increasing the number of simulations R also leads to a more accurate fit. This is because, like with increasing the sample size, the variation in the sample is reduced as a consequence of outliers having a lesser impact on the total sample.
- Some more explanations for these phenomena are: the **Law of Large Numbers**³, where larger sample sizes tend to produce a sample variance that more reliably approximates the expected gamma distribution; and the **Central Limit Theorem**^{4,5}, as it ensures that the mean of the sample variances converges to the population variance, with the distribution of sample variances becoming increasingly symmetric and less spread out as sample size increases.

³Investopedia. "Law of Large Numbers." Investopedia, Dotdash Meredith, <https://www.investopedia.com/terms/l/lawoflargenumbers.asp>. Accessed 9 Dec. 2024.

⁴Scribbr. "Central Limit Theorem | Formula, Definition & Examples." Scribbr, <https://www.scribbr.com/statistics/central-limit-theorem/>. Accessed 9 Dec. 2024.

⁵De Leon, Alexander. Lecture 9: Asymptotic Theory & Large-Sample Methods. Statistics 421, University of Calgary, Fall 2024.

2 Finding The Asymptotic Distribution of S^2

This section goes through the steps required to demonstrate that $S^2 \overset{a}{\sim} N\left(\sigma^2, \frac{\mu_4 - \sigma^4}{n}\right)$.

That is done by proving the equivalent statement: $\sqrt{n}(S^2 - \sigma^2) \xrightarrow{D} N(0, \mu_4 - \sigma^4)$, which is done using the **convergence limit theorems**, the Weak Law of Large Numbers(**WLLN**), the Central Limit Theorem (**CLT**), and **Slutsky's Theorem**⁶.

Rewriting $\sqrt{n}(S^2 - \sigma^2)$

To apply Slutsky's Theorem, $\sqrt{n}(S^2 - \sigma^2)$ must be rearranged: into parts that converge in accordance with the theorem. Recall that:

$$\begin{aligned} \sum_{i=1}^n (X_i - \mu)^2 &= \sum_{i=1}^n \left[(X_i - \bar{X}) + (\bar{X} - \mu) \right]^2 \\ &= \sum_{i=1}^n (X_i - \bar{X})^2 + 2 \sum_{i=1}^n (X_i - \bar{X})(\bar{X} - \mu) + \sum_{i=1}^n (\bar{X} - \mu)^2 \\ &= \sum_{i=1}^n (X_i - \bar{X})^2 + 2(\bar{X} - \mu) \left(\sum_{i=1}^n X_i - \sum_{i=1}^n \bar{X} \right) + n(\bar{X} - \mu)^2 \\ &= \sum_{i=1}^n (X_i - \bar{X})^2 + 2(\bar{X} - \mu)(n\bar{X} - n\bar{X}) + n(\bar{X} - \mu)^2 \\ &= \sum_{i=1}^n (X_i - \bar{X})^2 + n(\bar{X} - \mu)^2 \end{aligned}$$

Which means:

$$\begin{aligned} S^2 &= \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2 \\ &= \frac{1}{n-1} \left(\sum_{i=1}^n (X_i - \mu)^2 - n(\bar{X} - \mu)^2 \right) \\ &= \frac{n}{n-1} \left(\frac{1}{n} \sum_{i=1}^n (X_i - \mu)^2 - (\bar{X} - \mu)^2 \right) \end{aligned}$$

⁶De Leon, Alexander. Lecture 9: Asymptotic Theory & Large-Sample Methods. Statistics 421, University of Calgary, Fall 2024.

Now,

$$\begin{aligned}
\sqrt{n} (S^2 - \sigma^2) &= \sqrt{n} \left[\frac{n}{n-1} \left(\frac{1}{n} \sum_{i=1}^n (X_i - \mu)^2 - (\bar{X} - \mu)^2 \right) - \sigma^2 \right] \\
&= \sqrt{n} \left[\frac{n}{n-1} \left(\frac{1}{n} \sum_{i=1}^n (X_i - \mu)^2 - (\bar{X} - \mu)^2 - \frac{n-1}{n} \sigma^2 \right) \right] \\
&= \frac{n}{n-1} \left[\sqrt{n} \left(\frac{1}{n} \sum_{i=1}^n (X_i - \mu)^2 - (\bar{X} - \mu)^2 - \left(1 - \frac{1}{n}\right) \sigma^2 \right) \right] \\
&= \frac{n}{n-1} \left(\sqrt{n} \left[\frac{1}{n} \sum_{i=1}^n (X_i - \mu)^2 - \sigma^2 - (\bar{X} - \mu)^2 + \frac{1}{n} \sigma^2 \right] \right) \\
&= \frac{n}{n-1} \left(\sqrt{n} \left[\frac{1}{n} \sum_{i=1}^n (X_i - \mu)^2 - \sigma^2 \right] - \sqrt{n} (\bar{X} - \mu)^2 + \frac{1}{\sqrt{n}} \sigma^2 \right)
\end{aligned}$$

Showing Convergence of Each Term

From the CLT, $\sqrt{n} \left[\frac{1}{n} \sum_{i=1}^n (X_i - \mu)^2 - \sigma^2 \right] \xrightarrow{D} N(0, t^2)$ where $t^2 = \text{Var} \left[(X_i - \mu)^2 \right]$.

The asymptotic variance is from the fact that $\frac{1}{n} \sum_{i=1}^n (X_i - \mu)^2$ is the average of all $(X_i - \mu)^2$. Using the fact that each X_i come from a random sample,

$$\begin{aligned}
\text{Var} \left[(X_i - \mu)^2 \right] &= E \left[(X_i - \mu)^4 \right] - \left[E (X_i - \mu)^2 \right]^2 \\
&= E \left[(X_i - E[X_i])^4 \right] - \left[E (X_i - E[X_i])^2 \right]^2 \\
&= \mu_4 - \left[\text{Var} (X_i)^2 \right] \\
&= \mu_4 - [\sigma^2]^2 \\
&= \mu_4 - \sigma^4
\end{aligned}$$

Thus, $\sqrt{n} \left[\frac{1}{n} \sum_{i=1}^n (X_i - \mu)^2 - \sigma^2 \right] \xrightarrow{D} N(0, \mu_4 - \sigma^4)$

Consider,

$$\sqrt{n} (\bar{X} - \mu)^2 = \sqrt{n} (\bar{X} - \mu) (\bar{X} - \mu)$$

From the CLT,

$$\sqrt{n} (\bar{X} - \mu) \xrightarrow{D} N(0, \sigma^2)$$

And from the WLLN,

$$\begin{aligned} \bar{X} &\xrightarrow{P} \mu \\ (\bar{X} - \mu) &\xrightarrow{P} 0 \end{aligned}$$

From Slutsky's Theorem,

$$\sqrt{n} (\bar{X} - \mu) (\bar{X} - \mu) \xrightarrow{D} 0 \cdot N(0, \sigma^2)$$

$0 \cdot N(0, \sigma^2)$ is equal to the degenerate distribution degenerate at zero, which means $\sqrt{n} (\bar{X} - \mu) (\bar{X} - \mu) \xrightarrow{D} 0$. Then, from the sufficient condition for consistency, $\sqrt{n} (\bar{X} - \mu) (\bar{X} - \mu) \xrightarrow{P} 0$ is also true.

Now, $\frac{n}{n-1} \xrightarrow{P} 1$ because

$$\begin{aligned} \lim_{n \rightarrow +\infty} P \left(\left| \frac{n}{n-1} - 1 \right| < \epsilon \right) &= \lim_{n \rightarrow +\infty} P \left(\left| \frac{n}{n-1} - \frac{n-1}{n-1} \right| < \epsilon \right) \\ &= \lim_{n \rightarrow +\infty} P \left(\left| \frac{1}{n-1} \right| < \epsilon \right) \\ &= P \left(\left| \lim_{n \rightarrow +\infty} \frac{1}{n-1} \right| < \epsilon \right) \\ &= P(0 < \epsilon) \end{aligned}$$

Since ϵ is chosen to be any positive number, $0 < \epsilon$ is always true. $P(0 < \epsilon) = 1$ and by extension,

$$\lim_{n \rightarrow +\infty} P \left(\left| \frac{n}{n-1} - 1 \right| < \epsilon \right) = 1$$

. Which shows that $\frac{n}{n-1} \xrightarrow{P} 1$.

Also, $\frac{1}{\sqrt{n}}\sigma^2 \xrightarrow{P} 0$ since

$$\begin{aligned}\lim_{n \rightarrow +\infty} P\left(\left|\frac{1}{\sqrt{n}}\sigma^2 - 0\right| < \epsilon\right) &= \lim_{n \rightarrow +\infty} P\left(\left|\frac{1}{\sqrt{n}}\sigma^2\right| < \epsilon\right) \\ &= P\left(\left|\lim_{n \rightarrow +\infty} \frac{1}{\sqrt{n}}\sigma^2\right| < \epsilon\right) \\ &= P(0 < \epsilon)\end{aligned}$$

Since ϵ is chosen to be any positive number, $0 < \epsilon$ is always true. $P(0 < \epsilon) = 1$ and by extension,

$$\lim_{n \rightarrow +\infty} P\left(\left|\frac{1}{\sqrt{n}}\sigma^2 - 0\right| < \epsilon\right) = 1$$

. Which shows that $\frac{1}{\sqrt{n}}\sigma^2 \xrightarrow{P} 0$.

From the closure property under linearity I:

$$\begin{aligned}\sqrt{n}(\bar{X} - \mu)(\bar{X} - \mu) &\xrightarrow{P} 0 \\ \text{means} \\ -\sqrt{n}(\bar{X} - \mu)(\bar{X} - \mu) &\xrightarrow{P} 0 \\ \text{and} \\ -\sqrt{n}(\bar{X} - \mu)^2 &\xrightarrow{P} 0\end{aligned}$$

Then, from the closure property under linearity II,

$$\begin{aligned}\text{Since} \\ -\sqrt{n}(\bar{X} - \mu)^2 &\xrightarrow{P} 0 \text{ and } \frac{1}{\sqrt{n}}\sigma^2 \xrightarrow{P} 0 \\ \text{it means} \\ -\sqrt{n}(\bar{X} - \mu)^2 + \frac{1}{\sqrt{n}}\sigma^2 &\xrightarrow{P} 0\end{aligned}$$

Putting the Solution Together

Consider $\sqrt{n} (S^2 - \sigma^2)$ again,

$$\begin{aligned}\sqrt{n} (S^2 - \sigma^2) &= \frac{n}{n-1} \left(\sqrt{n} \left[\frac{1}{n} \sum_{i=1}^n (X_i - \mu)^2 - \sigma^2 \right] - \sqrt{n} (\bar{X} - \mu)^2 + \frac{1}{\sqrt{n}} \sigma^2 \right) \\ &= \frac{n}{n-1} \left(\sqrt{n} \left[\frac{1}{n} \sum_{i=1}^n (X_i - \mu)^2 - \sigma^2 \right] \right) + \frac{n}{n-1} \left(-\sqrt{n} (\bar{X} - \mu)^2 + \frac{1}{\sqrt{n}} \sigma^2 \right)\end{aligned}$$

Since $\frac{n}{n-1} \xrightarrow{P} 1$ and $\left(-\sqrt{n} (\bar{X} - \mu)^2 + \frac{1}{\sqrt{n}} \sigma^2 \right) \xrightarrow{P} 0$,

$\frac{n}{n-1} \left(-\sqrt{n} (\bar{X} - \mu)^2 + \frac{1}{\sqrt{n}} \sigma^2 \right) \xrightarrow{P} 0$ by the closure property under multiplication.

Finally, from Slutsky's theorem:

$$\begin{aligned}\sqrt{n} (S^2 - \sigma^2) &= \underbrace{\frac{n}{n-1}}_{\xrightarrow{P} 1} \underbrace{\left(\sqrt{n} \left[\frac{1}{n} \sum_{i=1}^n (X_i - \mu)^2 - \sigma^2 \right] \right)}_{\xrightarrow{D} N(0, \mu_4 - \sigma^4)} + \underbrace{\frac{n}{n-1} \left(-\sqrt{n} (\bar{X} - \mu)^2 + \frac{1}{\sqrt{n}} \sigma^2 \right)}_{\xrightarrow{P} 0} \\ &\xrightarrow{D} 0 + 1 \cdot N(0, \mu_4 - \sigma^4) \\ &= N(0, \mu_4 - \sigma^4)\end{aligned}$$

Hence, because $\sqrt{n} (S^2 - \sigma^2)$ has been shown to converge in distribution to $N(0, \mu_4 - \sigma^4)$,
 $S^2 \overset{a}{\sim} N\left(\sigma^2, \frac{\mu_4 - \sigma^4}{n}\right)$.

3 A Comparison Between the Exact and Asymptotic Distribution of S^2

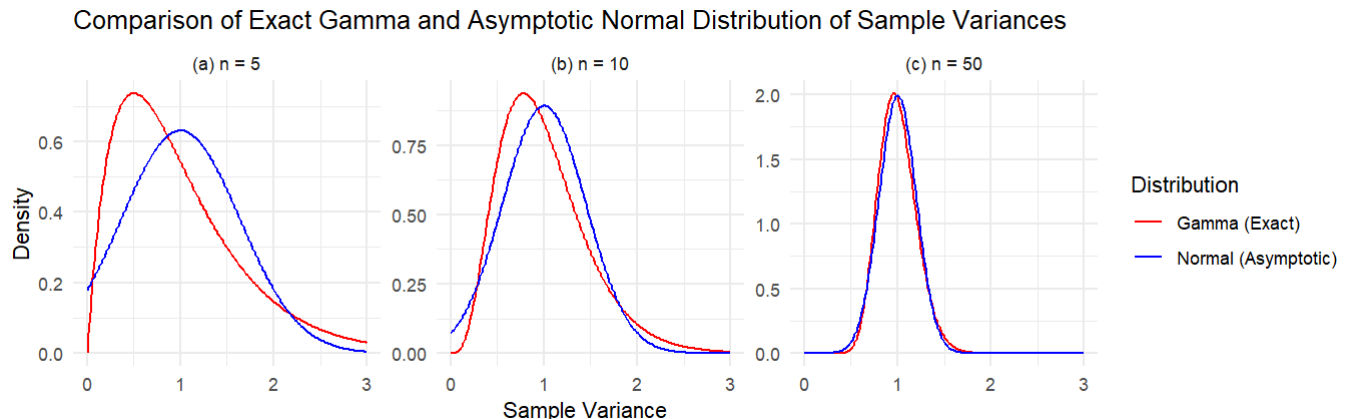


Figure 3: Exact vs. asymptotic distribution of S^2 for different sample sizes n^7

Observation from the Plots:

- **Sample Size $n=5$:** The normal distribution does not approximate the gamma distribution well. There is a noticeable difference in the peak and spread of the two distributions, with the gamma distribution showing higher kurtosis and the normal distribution being flatter and wider.
- **Sample Size $n=10$:** The fit between the normal and gamma distributions improves slightly. The peak positions are closer, although the gamma distribution still exhibits more pronounced tails.
- **Sample Size $n=50$:** The approximation is significantly better, with the peaks and shapes of the distributions aligning more closely. The normal distribution provides a good fit to the gamma distribution, particularly around the mode.

⁷See Appendix C for plot code

Analysis:

- **Impact of Sample Size (n):** As the sample size increases, the normal approximation improves substantially. This is expected due to the Central Limit Theorem (CLT), which implies that as n increases, the distribution of the sample variance S^2 , normalized by its mean and variance, will increasingly resemble the normal distribution. This is also supported by the properties of the gamma distribution, which approaches normality as its shape parameter increases (shape parameter for S^2 is $\frac{n-1}{2}$).
- **Effectiveness Across the Range of S^2 :** The normal approximation performs well around the mode of the gamma distribution but can deviate in the tails, particularly for smaller n . This deviation is more noticeable where the gamma distribution exhibits asymmetry or higher kurtosis than the normal distribution.
- **Improving the Normal Approximation:** For smaller sample sizes, the normal approximation may not be adequate, especially in applications requiring precision in the tail behavior (e.g., extreme value estimation or risk assessment). To improve the approximation:
 - Increase Sample Size: Using larger sample sizes generally enhances the approximation due to tighter CLT implications.
 - Transformation Techniques: Applying variance-stabilizing transformations to S^2 can sometimes yield distributions that are more symmetric and thus better approximated by the normal distribution. Techniques like the square root or logarithmic transformations could be considered, followed by an adjustment back to the original scale for interpretation.
 - Alternative Distributions: For critical applications, we can consider using parametric or non-parametric bootstrap methods to estimate the distribution of S^2 directly, rather than relying on asymptotic normality.

Conclusion

The normal approximation of the exact gamma sampling distribution of S^2 becomes more accurate with increasing sample size. While the approximation is generally good for practical purposes, especially for large n , care should be taken when making inferences based on this approximation for small sample sizes or when tail behavior is critical. Adjustments or alternative methods might be needed to address specific analytical requirements.

References

- [1] Investopedia. "Law of Large Numbers." Investopedia, Dotdash Meredith, <https://www.investopedia.com/terms/l/lawoflargenumbers.asp>. Accessed 9 Dec. 2024.
- [2] Scribbr. "Central Limit Theorem | Formula, Definition & Examples." Scribbr, <https://www.scribbr.com/statistics/central-limit-theorem/>. Accessed 9 Dec. 2024.
- [3] De Leon, Alexander. Lecture 9: Asymptotic Theory & Large-Sample Methods. Statistics 421, University of Calgary, Fall 2024.

Appendix

A Python code for Question 1a

```
import numpy as np

# Define the number of simulations and sample sizes
R = 10000
sample_sizes = [5, 10, 50]
mu = 0
sigma_squared = 1

# Function to calculate  $S^2$  for a sample
def calculate_s_squared(sample):
    n = len(sample)
    sample_mean = np.mean(sample)
    return np.sum((sample - sample_mean)**2) / (n - 1)

# Monte Carlo Simulation to calculate  $S^2$ 
results = []
for n in sample_sizes:
    s_squared_values = np.array(
        [calculate_s_squared(np.random.normal(mu, np.sqrt(sigma_squared), n)) for _ in range(R)])
    monte_carlo_mean = np.mean(s_squared_values)
    monte_carlo_sd = np.std(s_squared_values, ddof=1)
    exact_mean = sigma_squared # Since  $S^2$  is an unbiased estimator of  $\sigma^2$ 
    exact_sd = np.sqrt(2 * sigma_squared**2 / (n - 1)) # SD for Gamma distribution
    results.append((n, monte_carlo_mean, exact_mean, monte_carlo_sd, exact_sd))

print(results)
```

Output:

$R = 10000$:

```
[(5, 0.9955727922184622, 1,  
 0.7025651471777558, 0.7071067811865476),  
(10, 0.9951928419108039, 1,  
0.47304737346719483, 0.4714045207910317),  
(50, 0.997165487152457, 1,  
0.20056637039354624, 0.20203050891044214)]
```

$R = 100000$:

```
[(5, 1.000376421951266, 1,  
0.7057581182000536, 0.7071067811865476),  
(10, 0.999679560777648, 1,  
0.47255480399127264, 0.4714045207910317),  
(50, 0.9999017650243442,  
1, 0.2016684592200472, 0.20203050891044214)]
```

B R code for Question 1b

```
#Load necessary libraries
library(ggplot2)

#Function to simulate sample variances
simulate_sample_variances <- function(n, R) {
  s2_values <- numeric(R)
  for (i in 1:R){
    samples <- rnorm(n,mean=0,sd=1)
    s2_values[i] <- var(samples)
  }
  return(s2_values)
}

#Function to get letter for plot title
get_letter <- function(n){
  if(n == 5){
    return("(a)")
  }
  else if(n == 10){
    return("(b)")
  }
  else{
    return("(c)")
  }
}
```

Plot code continued

```
#Parameters
sample_sizes <- c(5,10,50)
num_simulations <- 10000

#Setup the plotting window
par(mfrow = c(3,1))

#Loop through each sample size
for(n in sample_sizes){
  #Simulate sample variances
  s2_values <- simulate_sample_variances(n,num_simulations)

  #Calculate mean and SD for gamma parameters
  alpha <- (n-1)/2
  scale <- 2/(n-1)

  #Create histogram and density plot
  hist(s2_values, freq=FALSE, breaks=30,
       main = paste(get_letter(n)," Sample Variance with n =", n),
       xlab = "Sample Variance", ylab= "Density", col="lightblue",)
  curve(dgamma(x,shape=alpha,scale=scale), add=TRUE, col="red",lwd=2, bty="n")
  abline(v=mean(s2_values),col="black", lty=2, lwd = 2)
  legend("topright",legend=c("Exact PDF","Variance Mean"),
       col=c("red","black"),lty=1:2, cex = 0.75)
}
```

C R code for Question 3

```
#Include libraries
library(ggplot2)
library(dplyr)

#Set parameters
sigma2 <- 1 #Variance of the normal distribution
mu4 <- 3 * sigma2^2 #Fourth central moment of the normal distribution

#Function to generate data frames of density values for gamma and normal distributions
generate_densities <- function(n){
  alpha <- (n-1)/2
  beta <- 2 * sigma2 / (n-1)
  x <- seq(0,3,length.out=200)
  gamma_dens <- dgamma(x,shape=alpha,scale=beta)
  normal_dens <- dnorm(x,mean=sigma2, sd = sqrt((mu4 - sigma2^2)/n))

  data.frame(
    x=x,
    Density = c(gamma_dens,normal_dens),
    Distribution = c(rep('Gamma (Exact)', length(x)), rep('Normal (Asymptotic)', length(x))),
    Sample_Size = paste(get_letter(n),"n =",n)
  )
}

#Function to get letter for plot title
get_letter <- function(n){
  if(n == 5){
    return("(a)")
  }
  else if(n == 10){
    return("(b)")
  }
  else{
    return("(c)")
  }
}
```

Plot code continued

```
#Generate densities for different n
all_densities <- bind_rows(generate_densities(5),generate_densities(10),generate_densities(50))

#Plotting
ggplot(all_densities, aes(x=x,y=Density,color=Distribution)) +
  geom_line() +
  facet_wrap(~Sample_Size, scales = 'free_y') +
  labs(title = 'Comparison of Exact Gamma and Asymptotic Normal Distribution of Sample Variances',
       x = 'Sample Variance',
       y = 'Density') +
  theme_minimal() +
  scale_color_manual(values = c('red','blue'))
```